Pool Ball Detection and Recognition

Mahmoud Hatem Alaskandrani

Trinity College Dublin / The American University in Cairo

November 8, 2026

Abstract

This report documents the experimental development of a two-stage computer vision pipeline for localising and recognising pool balls. Stage 1 tested a green-channel boost with adaptive thresholding, fixed HSV and single-channel lightness thresholding, and finally a successful multi-channel CIELAB edge-fusion detector with a dynamic L-channel pocket mask and geometric cushion constraints. Stage 2 tested K-Means colour clustering and finally adopted one-dimensional hue and saturation histograms with a reduced circular sampling mask and a 30/70 weighting scheme. The final pipeline achieves Stage 1 precision 0.785, recall 0.933 and F1-score 0.853, together with Stage 2 recognition accuracy 70.2% across the six test tables.

1 Introduction

The task was to localise every pool ball on a rectified table image and then assign each surviving detection to one of the sixteen known balls. The six test images contain strong shadows, uneven lighting, tightly packed racked balls, and a mixture of solids and stripes. I developed the final system through a sequence of explicit experiments, and I retained a detailed error analysis because the remaining failures reveal the limits of the chosen detectors and descriptors.

2 Stage 1: Ball Detection

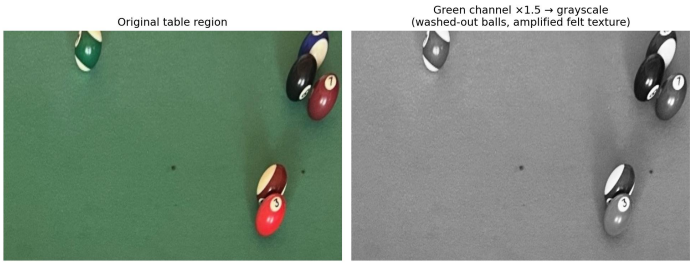
2.1 Perspective rectification

Each input photograph is first mapped by a planar homography from the four measured cushion-to-cushion corners into a 1434×683 metric table frame. The destination dimensions match the physical $1434\,\text{mm} \times 683\,\text{mm}$ playing area, so one transformed pixel corresponds to one millimetre. This metric frame is used by all later stages.

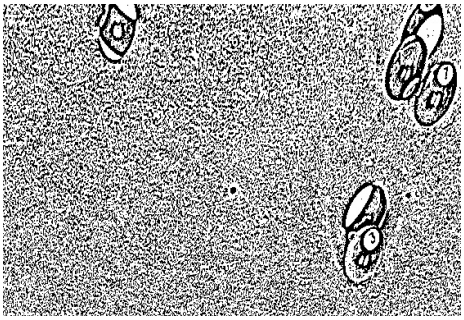
2.2 Failed green-channel boost and adaptive thresholding

To solve the problem of missing green balls, I initially hypothesised that isolating and boosting the RGB green channel by a factor of 1.5, followed by adaptive Gaussian thresholding, would force the balls to pop against the felt. The adaptive threshold used a local block size of 15 and a constant of 2. This failed because boosting the green channel

equally boosted the green table, and adaptive thresholding amplified the woven texture of the felt into severe noise. Figure 1 shows the original region and the green-boosted grayscale image, together with the resulting adaptive threshold map. The ball boundaries are washed out while the felt texture dominates the edge map.



(a) Original region versus green-channel boost.

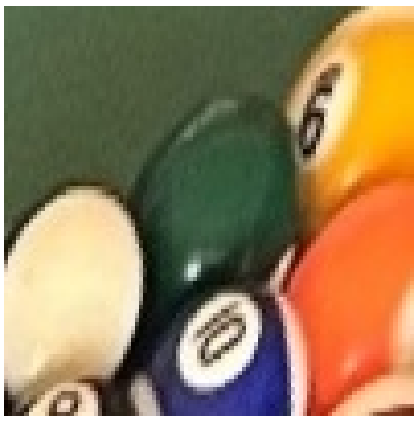


(b) Adaptive Gaussian threshold map: severe felt texture noise.

Figure 1: The green-channel boost and adaptive-threshold experiment. The boosted image washes out ball features, and the threshold map is dominated by felt texture rather than ball boundaries.

2.3 Failed HSV and single-channel lightness thresholding

I then attempted to isolate the felt directly in HSV space with a fixed hue range of 35–85, saturation range 40–255, and value range 40–255. This also failed for the green object balls, particularly balls 6 and 14, because their hue and saturation overlap almost completely with the green felt. Figure 2 shows a 100×100 region around the green ball and the resulting binary mask. In the mask, the green ball vanishes into the white felt instead of producing a distinct foreground blob. A single-channel lightness threshold produced the complementary failure: shadows fragmented the ball boundaries and the threshold could not separate dark balls from the table. Both results motivated a multi-channel approach.



(a) Original region around the green ball.



(b) Fixed HSV threshold: the green ball merges into the felt.

Figure 2: Visual evidence for the failure of fixed HSV thresholding. The green ball and the green felt are not separable with static hue and saturation limits.

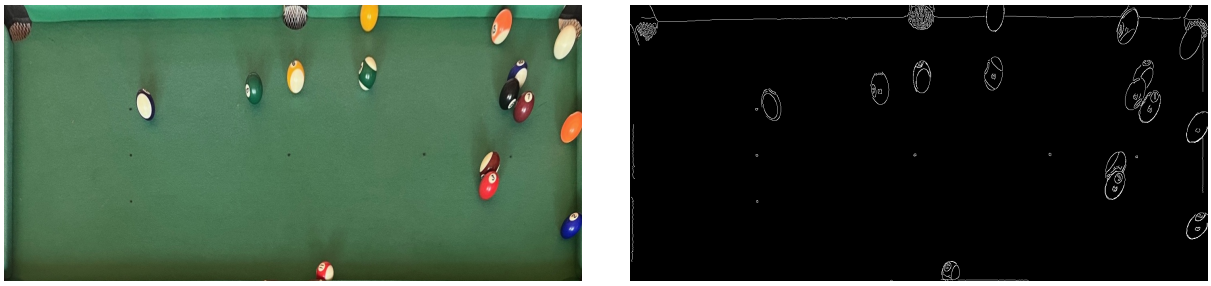
2.4 Successful multi-channel CIELAB edge fusion

I realised that the solution required looking at colour boundaries, not just brightness. I split the rectified image into the CIELAB colour space and applied Gaussian smoothing with a 5×5 kernel and a standard deviation of 1.2 to all three channels. I then ran Canny edge detection independently on each channel. I used thresholds of 45 and 95 for the lightness channel L, and more sensitive thresholds of 35 and 85 for the chromatic channels a and b. The three binary edge maps were combined with a logical OR to produce a highly robust gradient map that captured subtle colour shifts as well as luminance boundaries.

Key Idea: LAB Multi-Channel Edge Fusion

The lightness channel supplies the strong outer silhouette of each ball, while the chromatic channels supply the colour boundaries that remain visible when a ball is dark, shadowed, or similar in brightness to the felt. Fusing the three edge maps produces a gradient image that is markedly more complete than any single-channel edge map.

Figure 3 shows the rectified Table 4 image and the corresponding fused edge map used by the circular detector.



(a) Rectified Table 4. (b) Multi-channel LAB edge fusion map.

Figure 3: Rectification and the successful fused edge map for Table 4.

2.5 Dynamic pocket mask and geometric constraints

The fused edge map is passed to the Hough Circle Transform with a minimum centre separation of 40 pixels and a ball-radius range of 19 to 24 pixels. I initially rejected every candidate within a blind 95-pixel radius of the six theoretical pockets, but visual debugging showed that this removed genuine balls near the corners, including the bright cue ball beside the top-right pocket of Table 4. I therefore replaced the blind radius with a dynamic pocket mask built from L-channel extremes.

For each of the six pocket centres, I examined a 120×120 pixel region on the raw lightness channel and thresholded two complementary structures: the dark pocket holes at $L < 35$ and the bright metallic brackets at $L > 200$. The local mask was restricted to a 55-pixel radius around the pocket centre so that a bright ball near a pocket would not be mistaken for a metallic bracket. The six local masks were combined into a global binary mask and dilated with a 13×13 circular kernel to cover the immediate pocket lip. A Hough candidate is now rejected only when its centre lies on the dynamic pocket mask. The 22-pixel cushion exclusion remains active.

Figure 4 compares the old blind zone with the new mask at the top-right pocket of Table 4. The old red circle reaches into the cue ball's neighbourhood, while the green dynamic mask covers only the dark hole and bracket. The cue ball is therefore spared and is recovered by the final detector.

Table 4 top-right pocket: blind radius vs dynamic pocket mask

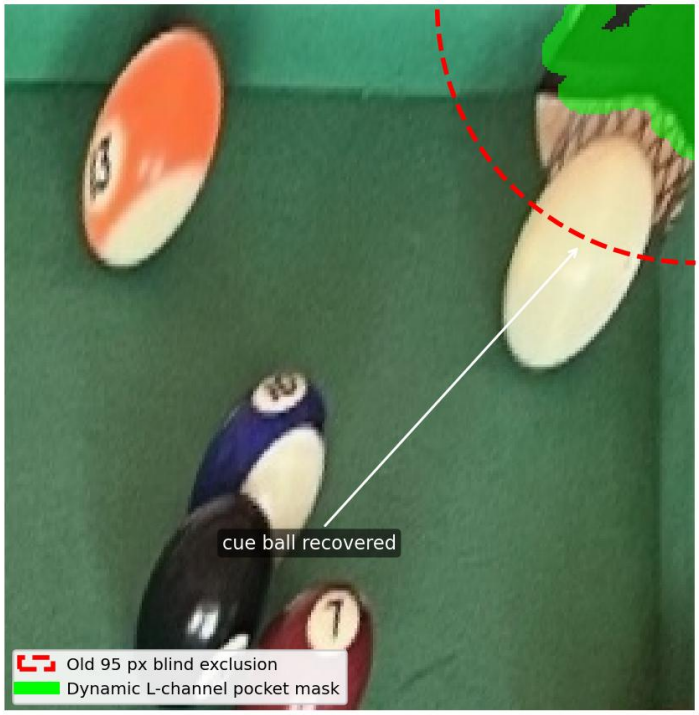


Figure 4: Top-right pocket of Table 4. The dashed red circle is the old blind 95-pixel exclusion; the green region is the dynamic L-channel pocket mask. The cue ball is no longer removed.

3 Stage 2: Ball Recognition and Classification

3.1 Failed K-Means clustering

To differentiate striped from solid balls, I first attempted to use K-Means clustering with $K = 2$ on the pixels inside each ball mask to extract two dominant colours, then compared those colours with the reference balls using Euclidean distance. This proved computationally heavy and unreliable because the visibility of the white stripe depends on the physical rotation of the ball. Figure 5 shows the result on a solid blue ball. The two clusters split the number and the specular reflections rather than producing a stable stripe-versus-solid descriptor.

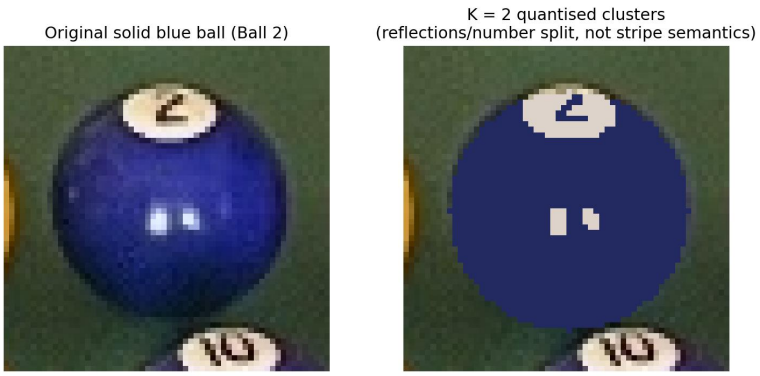


Figure 5: K-Means ($K = 2$) applied to a solid blue ball. The clustering splits the number and reflections, not a rotationally stable stripe/solid feature.

3.2 Successful saturation-dominant histograms

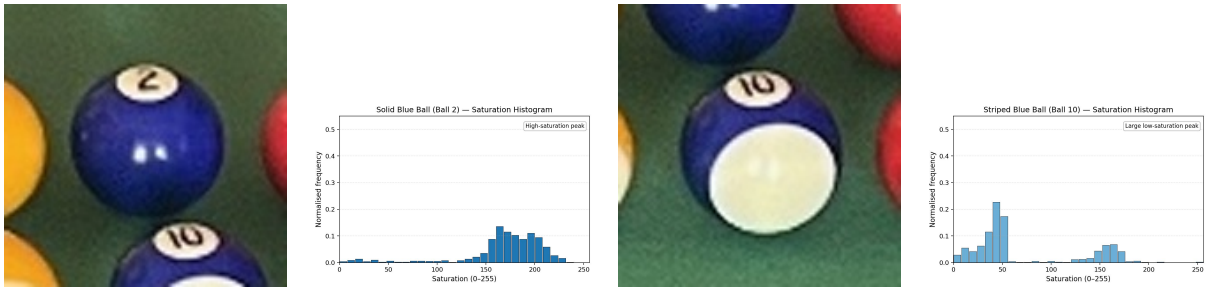
I pivoted to one-dimensional histograms. I extracted a hue histogram with 30 bins and a saturation histogram with 32 bins from a reduced circular sampling mask of radius 16 pixels. The reduced mask prevents a slightly off-centre detection from admitting felt, shadow, or corner pixels into the descriptor. I then compared test histograms with the sixteen reference histograms using the Bhattacharyya distance. Because a white stripe drastically lowers a ball's average saturation regardless of rotation, I applied a custom weighting system: 30% weight to hue, 70% weight to saturation, and 0% weight to brightness.

Saturation-Dominant Distance

$$d = 0.30\, d_{\text{hue}} + 0.70\, d_{\text{sat}}. \tag{1}$$

A detection whose weighted distance exceeds 0.80 is rejected as a non-ball or as an unreliable observation.

Figure 6 compares the solid blue ball 2 with the striped blue ball 10. The solid ball peaks at high saturation, whereas the striped ball has a pronounced low-saturation peak caused by its white stripe. This shift is independent of the ball's physical rotation and provides a mathematically robust feature for distinguishing stripes from solids.



(a) Ball 2 crop. (b) Solid histogram. (c) Ball 10 crop. (d) Striped histogram.

Figure 6: Solid versus striped blue balls and their saturation histograms. The white stripe shifts the striped ball's distribution strongly toward the low-saturation end, motivating the 70% saturation weighting.

4 Results and Error Analysis

4.1 Stage 1: Detection Results

Table 1 reports the final Stage 1 metrics. The dynamic pocket mask recovers the cue and other balls near pockets, raising recall to 0.933. Overall precision is 0.785 and the F1-score is 0.853.

Table 1: Stage 1 detection metrics after dynamic pocket masking.

Image	TP	FP	FN	Precision	Recall	F1-score
Table 1	16	4	0	0.800	1.000	0.889
Table 2	16	5	0	0.762	1.000	0.865
Table 3	16	2	0	0.889	1.000	0.941
Table 4	14	3	1	0.824	0.933	0.875
Table 5	12	5	2	0.706	0.857	0.774
Table 6	10	4	3	0.714	0.769	0.741
Overall	84	23	6	0.785	0.933	0.853

4.2 Stage 2: Recognition Results

Table 2 reports recognition accuracy for the true-positive detections. Across all six tables, 59 of 84 correctly localised balls receive the correct label, giving an overall recognition accuracy of 70.2%. Figure 7 shows the final labelled output for Table 4, including the recovered cue ball.

Table 2: Stage 2 recognition accuracy on correctly localised balls.

Image	Correct labels	True positives	Accuracy
Table 1	11	16	0.688
Table 2	10	16	0.625
Table 3	11	16	0.688
Table 4	12	14	0.857
Table 5	8	12	0.667
Table 6	7	10	0.700
Overall	59	84	0.702

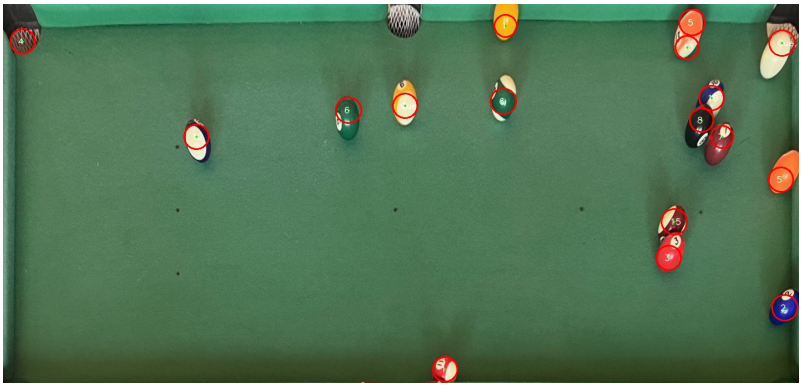


Figure 7: Final labelled output for Table 4 after dynamic pocket masking and Stage 2 classification.

4.3 Error Analysis

The remaining Stage 1 errors are dominated by two failure modes. Of the 23 false positives, 14 are cushion or rail artefacts and 9 are duplicate responses on already-detected balls. The cushion artefacts arise because the wooden rail trim and the cushion seam produce short circular arcs that survive the multi-channel edge map. The duplicate responses occur when a ball has a strong internal marking, such as a number or a stripe boundary, so that the circular accumulator produces a second centre slightly offset from the true one. Of the 6 false negatives, 4 are balls whose centres still fall inside the 22-pixel cushion margin or close to a dynamic pocket mask, and 2 are balls under extreme shadow or very low local contrast. No false negatives remain in Tables 1–3; the residual misses are concentrated in Tables 4–6, where the balls are closest to the rails and pockets.

Stage 2 also has a clear failure structure. Of the 84 true-positive detections, 59 receive the correct label and 25 are misclassified, giving an accuracy of 70.2%. Nine of the misclassifications are direct solid/stripe confusions of the same base colour (for example, a solid blue and a blue stripe). The remaining sixteen are broader histogram-overlap confusions between different colours, especially low-saturation balls and balls whose visible stripe or number changes the hue and saturation distribution. The saturation weighting improves the stripe/solid separation, but a global histogram still discards the spatial arrangement of the stripe and number, so a rotated stripe can resemble a different solid ball.

5 Discussion

The multi-channel LAB edge-fusion detector outperformed the green-boost, adaptive-threshold, fixed-HSV, and single-channel lightness experiments. The lightness channel supplies the outer silhouette, while the a and b channels contribute colour boundaries that remain visible when a ball is dark, shadowed, or similar in brightness to the felt. The dynamic pocket mask is a clear improvement over the blind radius: it removes the true pocket structures while sparing nearby balls, which raises recall from 0.867 to 0.933. The cost is a lower precision of 0.785, because rail and duplicate responses are no longer removed by the over-large blind zone. The resulting Stage 1 F1-score is 0.853, still above the previous single-channel baseline of approximately 0.831.

Recognition is limited mainly by the ambiguity between similarly coloured solids and stripes. The saturation-dominant histogram improves on K-Means, but the residual errors show that a global hue and saturation distribution cannot fully capture the spatial arrangement of the stripe and number. Higher-resolution descriptors that preserve spatial structure, or a dedicated stripe detector, would be the natural next step.

6 Conclusion

I replaced the failing HSV and brightness-threshold designs with a multi-channel CIELAB edge-fusion detector, and I replaced the unstable K-Means recognition with a 30/70 hue–saturation histogram descriptor. The dynamic L-channel pocket mask recovers balls that the blind radius excluded, lifting Stage 1 recall to 0.933 and F1 to 0.853. Stage 2 recognition accuracy is 0.702. The error analysis shows that the remaining failures are concentrated in rail and duplicate responses for detection and in stripe/solid and low-saturation colour overlap for recognition.